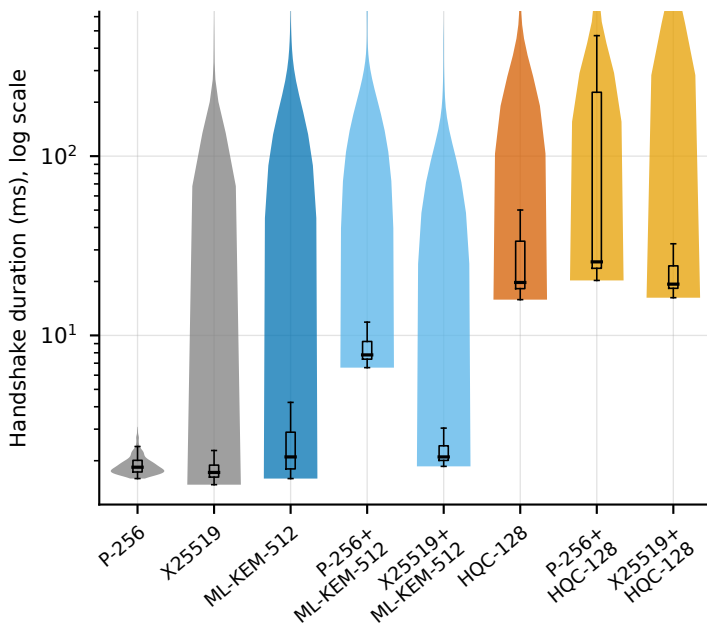
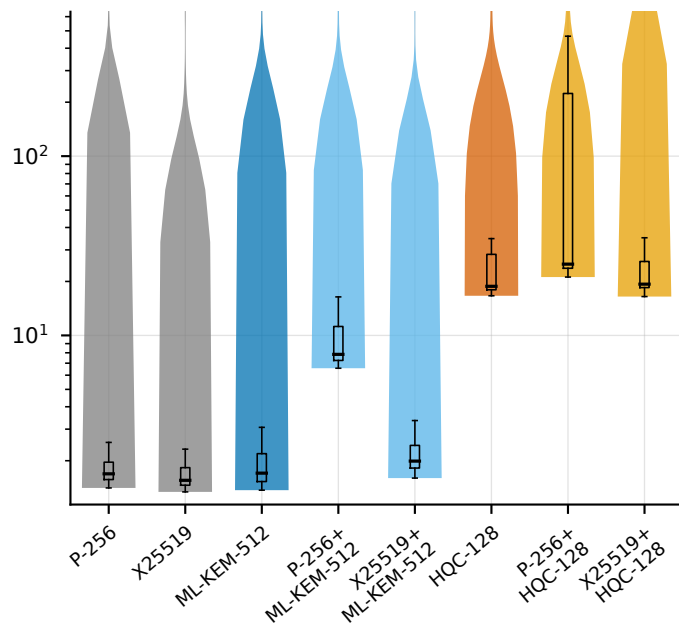


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	1.92	13.2	3.09	105.8	622.4	13.23k
X25519	57.5	610.1	6.59k	71.4	396.3	3.15k
ML-KEM-512	157.0	287.6	4.31k	142.4	423.8	7.85k
P-256+ML-KEM-512	181.4	265.8	3.27k	189.9	299.4	7.64k
X25519+ML-KEM-512	65.7	397.3	2.30k	111.5	403.3	6.80k
HQC-128	249.6	261.5	8.67k	246.6	242.5	4.32k
P-256+HQC-128	315.6	256.9	13.41k	323.3	227.9	7.66k
X25519+HQC-128	232.0	560.8	26.42k	337.3	455.8	30.75k